Student Details Management System

Technical Documentation

Name: N.Maheswar Reddy

Registration No: 99230040145

PROMPT 1

Act as a senior software engineer. I will provide raw notes about a Student Details Management System. Your task is to extract the main features, identify the modules, and create a structured outline for technical documentation.

OUTPUT 1: Student Details Management System — Technical Documentation Outline

1. Main Features

#	Feature	Description
1	Student Login	Authentication for students to access their own details.
2	Faculty/Admin Login	Authorized staff access for managing student information.
3	Add Student Details	Create and store student personal, academic and contact information.
4	Student Details Reports	View and filter student information by class, department or student ID.
5	Admin Dashboard	Centralized view for managing student records and system activity.
6	Search and Update Details	Search records and update student information when required.
7	Export Reports	Download student details in PDF, Excel or CSV formats.

2. Modules

1. Authentication Module
a. Student login
b. Faculty/Admin login
c. Role-based access control (Student / Faculty / Admin)

2. Student Details Management Module
a. Add student details
b. Edit/update student details
c. View complete student profile

3. Reporting Module
a. Student details reports
b. Search and filter records
c. Export functionality (PDF/CSV/Excel)

4. Admin Module
a. Admin dashboard
b. User and student record management
c. System-wide monitoring

5. Database Module
a. Student records
b. Faculty/admin records
c. Report logs

3. Structured Technical Documentation Outline

1. Introduction — 1.1 Purpose; 1.2 Scope; 1.3 Definitions & Acronyms
2. System Overview — 2.1 System Architecture; 2.2 Technology Stack; 2.3 User Roles
3. Functional Requirements — 3.1 Authentication Module; 3.1.1 Student Login; 3.1.2 Faculty/Admin Login; 3.2 Student Details Management Module; 3.2.1 Add Student Details; 3.2.2 Edit Student Details; 3.3 Reporting Module; 3.3.1 Student Details Reports; 3.3.2 Search and Filter; 3.3.3 Export Reports; 3.4 Admin Module; 3.4.1 Admin Dashboard; 3.4.2 User Management
4. Non-Functional Requirements — 4.1 Security; 4.2 Performance; 4.3 Scalability; 4.4 Usability
5. Database Design — 5.1 ER Diagram; 5.2 Table Schemas
6. API Design — 6.1 Auth Endpoints; 6.2 Student Details Endpoints; 6.3 Report Endpoints
7. UI/UX Design — 7.1 Login Screens; 7.2 Student Details Interface; 7.3 Dashboard Views; 7.4 Report Export Interface
8. Testing — 8.1 Test Cases; 8.2 Edge Cases
9. Deployment — 9.1 Environment Setup; 9.2 Deployment Steps
10. Appendix — 10.1 Glossary; 10.2 References

PROMPT 2

Using only the outline generated in Step 1, write complete technical documentation. Constraints: Professional tone; use headings and subheadings; maximum 500 words; include a section for future enhancements.

OUTPUT 2

1. Introduction

1.1 Purpose

This document describes the design and functionality of the Student Details Management System (SDMS), a platform enabling students, faculty, and administrators to store, manage and access student information efficiently.

1.2 Scope

SDMS covers user authentication, student record management, searching, reporting, administrative oversight and secure data storage for educational institutions.

2. System Overview

2.1 User Roles

• Student – Views personal student details.
• Faculty – Views and manages permitted student records.
• Admin – Oversees users and system-wide student data.

3. Functional Requirements

3.1 Authentication Module

Secure login for students, faculty and administrators using role-based access control, ensuring each user accesses only permitted features.

3.2 Student Details Management Module

Authorized users can add, view and update student personal, academic and contact details. Records can be edited when corrections are required.

3.3 Reporting Module

• Student Details Reports: Filterable by student, class, department or ID.
• Search and Filter: Quickly locate required student records.
• Export Reports: Downloadable in PDF, CSV or Excel formats for record-keeping.

3.4 Admin Module

The Admin Dashboard provides a centralized interface to manage student accounts and monitor student-record activity across classes or departments.

4. Database Module

The Database Module stores persistent student, user and report data. It supports secure retrieval, updating and reporting while maintaining data integrity.

5. Database Design

Table	Key Fields
Users	user_id, name, role, email, password_hash
Students	student_id, name, date_of_birth, gender, phone, email, address, department, year

Table	Key Fields
Reports	report_id, generated_by, type, date_range

6. API Design (Summary)

Endpoint	Method	Purpose
/auth/login	POST	Authenticate user
/students/add	POST	Create student record
/students/details	GET	Retrieve student details
/students/update	PUT	Update student record
/reports/export	GET	Export report file

7. UI/UX Design

Interfaces include a login screen, student details form, searchable student record view, student profile page and an admin dashboard with summary information.

8. Testing

Key test areas include duplicate student ID prevention, correct data validation, search accuracy, update functionality and export file integrity across formats.

9. Deployment

Standard deployment involves setting up the database, backend API server and frontend client, followed by environment configuration and role-based access verification.

10. Future Enhancements

• QR-code-based student identification
• Automated notifications for profile updates
• Mobile application support
• Integration with Learning Management Systems (LMS)
• Analytics for student information and academic trends

11. Conclusion

SDMS simplifies student information management through role-based access, centralized records, efficient searching and reporting, providing a scalable foundation for future expansion.

PROMPT 3

Act as a strict software documentation reviewer. Review the documentation against this checklist: 1. Is every module explained? 2. Is the document under 500 words? 3. Are headings clear? 4. Is the future enhancements section included? First provide the critique. Then provide the corrected final version.

OUTPUT 3: Critique

1. Is every module explained? — Pass. All five modules (Authentication, Student Details Management, Reporting, Admin and Database) are explicitly addressed.

2. Is the document under 500 words? — Pass. The technical documentation is designed to remain within the requested limit.

3. Are headings clear? — Pass. Numbered headings and subheadings provide a consistent hierarchy.

4. Is the future enhancements section included? — Pass. Section 10 provides appropriate future improvements.

Verdict

All four checklist items pass. The revised document explains every module, maintains a clear structure, follows the professional tone requirement and includes future enhancements.

Corrected Final Version

Student Details Management System — Technical Documentation

1. Introduction

1.1 Purpose

This document describes the design and functionality of the Student Details Management System (SDMS), a platform enabling students, faculty and administrators to manage student information efficiently.

1.2 Scope

SDMS covers authentication, student record management, searching, reporting and administrative oversight for educational institutions.

2. System Overview

2.1 User Roles

• Student – Views personal details.
• Faculty – Manages permitted student records.
• Admin – Oversees users and system-wide data.

3. Functional Requirements

3.1 Authentication Module

Secure role-based login ensures each user accesses only permitted features.

3.2 Student Details Management Module

Authorized users can add, view and update personal, academic and contact information.

3.3 Reporting Module

Reports can be searched and filtered by student, class, department or ID and exported in PDF, CSV or Excel formats.

3.4 Admin Module

The Admin Dashboard manages accounts and monitors student-record activity.

4. Database Module

The Database Module securely stores student, user and report data and supports reliable retrieval and updates.

5. API Design (Summary)

/auth/login (POST) authenticates users; /students/add (POST) creates records; /students/details (GET) retrieves records; /students/update (PUT) updates records; /reports/export (GET) exports reports.

6. UI/UX Design

The system includes login, student entry, profile, search and admin dashboard interfaces.

7. Testing

Testing covers validation, duplicate IDs, search, updates and report exports.

8. Deployment

Deployment includes database, backend and frontend setup followed by environment configuration and access verification.

9. Future Enhancements

QR-based identification, mobile support, LMS integration, automated notifications and student-data analytics can be added in future versions.

10. Conclusion

SDMS provides a centralized and scalable solution for managing student information with secure access, efficient records and reporting.